Walmart Sales

March 14, 2026

1 Walmart Sales Project

[1]:
```
import pandas as pd
import numpy as np
```

[2]:
```
df = pd.read_csv("Walmart_Sales.csv")
```

[3]:
```
df
```

[3]:
```
      Store        Date  Weekly_Sales  Holiday_Flag  Temperature  Fuel_Price  \
0         1  05-02-2010    1643690.90             0        42.31       2.572
1         1  12-02-2010    1641957.44             1        38.51       2.548
2         1  19-02-2010    1611968.17             0        39.93       2.514
3         1  26-02-2010    1409727.59             0        46.63       2.561
4         1  05-03-2010    1554806.68             0        46.50       2.625
…       …          …           …           …           …          …
6430     45  28-09-2012     713173.95             0        64.88       3.997
6431     45  05-10-2012     733455.07             0        64.89       3.985
6432     45  12-10-2012     734464.36             0        54.47       4.000
6433     45  19-10-2012     718125.53             0        56.47       3.969
6434     45  26-10-2012     760281.43             0        58.85       3.882

             CPI  Unemployment
0     211.096358         8.106
1     211.242170         8.106
2     211.289143         8.106
3     211.319643         8.106
4     211.350143         8.106
…          …           …
6430  192.013558         8.684
6431  192.170412         8.667
6432  192.327265         8.667
6433  192.330854         8.667
6434  192.308899         8.667

[6435 rows x 8 columns]
```

[4]:
```
df.head()
```

[4]:
```
   Store        Date  Weekly_Sales  Holiday_Flag  Temperature  Fuel_Price  \
0      1  05-02-2010    1643690.90             0        42.31       2.572
1      1  12-02-2010    1641957.44             1        38.51       2.548
2      1  19-02-2010    1611968.17             0        39.93       2.514
3      1  26-02-2010    1409727.59             0        46.63       2.561
4      1  05-03-2010    1554806.68             0        46.50       2.625

          CPI  Unemployment
0  211.096358         8.106
1  211.242170         8.106
2  211.289143         8.106
3  211.319643         8.106
4  211.350143         8.106
```

[5]:
```
df.tail()
```

[5]:
```
      Store        Date  Weekly_Sales  Holiday_Flag  Temperature  Fuel_Price  \
6430     45  28-09-2012     713173.95             0        64.88       3.997
6431     45  05-10-2012     733455.07             0        64.89       3.985
6432     45  12-10-2012     734464.36             0        54.47       4.000
6433     45  19-10-2012     718125.53             0        56.47       3.969
6434     45  26-10-2012     760281.43             0        58.85       3.882

             CPI  Unemployment
6430  192.013558         8.684
6431  192.170412         8.667
6432  192.327265         8.667
6433  192.330854         8.667
6434  192.308899         8.667
```

[6]:
```
df.shape
```

[6]:
```
(6435, 8)
```

[8]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6435 entries, 0 to 6434
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Store         6435 non-null   int64
 1   Date          6435 non-null   object
 2   Weekly_Sales  6435 non-null   float64
 3   Holiday_Flag  6435 non-null   int64
```

```
 4   Temperature   6435 non-null   float64
 5   Fuel_Price    6435 non-null   float64
 6   CPI           6435 non-null   float64
 7   Unemployment  6435 non-null   float64
dtypes: float64(5), int64(2), object(1)
memory usage: 402.3+ KB
```

[9]:
```
df.describe
```

[9]:
```
<bound method NDFrame.describe of       Store        Date  Weekly_Sales
Holiday_Flag  Temperature  Fuel_Price  \
0         1  05-02-2010    1643690.90             0        42.31       2.572
1         1  12-02-2010    1641957.44             1        38.51       2.548
2         1  19-02-2010    1611968.17             0        39.93       2.514
3         1  26-02-2010    1409727.59             0        46.63       2.561
4         1  05-03-2010    1554806.68             0        46.50       2.625
…       …          …            …            …           …          …
6430     45  28-09-2012     713173.95             0        64.88       3.997
6431     45  05-10-2012     733455.07             0        64.89       3.985
6432     45  12-10-2012     734464.36             0        54.47       4.000
6433     45  19-10-2012     718125.53             0        56.47       3.969
6434     45  26-10-2012     760281.43             0        58.85       3.882

             CPI  Unemployment
0     211.096358         8.106
1     211.242170         8.106
2     211.289143         8.106
3     211.319643         8.106
4     211.350143         8.106
…           …           …
6430  192.013558         8.684
6431  192.170412         8.667
6432  192.327265         8.667
6433  192.330854         8.667
6434  192.308899         8.667

[6435 rows x 8 columns]>
```

[10]:
```
df.isnull().sum()
```

[10]:
```
Store           0
Date            0
Weekly_Sales    0
Holiday_Flag    0
Temperature     0
Fuel_Price      0
CPI             0
```

```
Unemployment    0
dtype: int64
```

[13]:
```python
print(df.columns)
```

```
Index(['Store', 'Date', 'Weekly_Sales', 'Holiday_Flag', 'Temperature',
       'Fuel_Price', 'CPI', 'Unemployment'],
      dtype='object')
```

2 Data Exploration (EDA)

3 1. Highest sales store

[14]:
```python
df.groupby("Store")["Weekly_Sales"].sum().sort_values(ascending=False)
```

[14]:
```
Store
20    3.013978e+08
4     2.995440e+08
14    2.889999e+08
13    2.865177e+08
2     2.753824e+08
10    2.716177e+08
27    2.538559e+08
6     2.237561e+08
1     2.224028e+08
39    2.074455e+08
19    2.066349e+08
31    1.996139e+08
23    1.987506e+08
24    1.940160e+08
11    1.939628e+08
28    1.892637e+08
41    1.813419e+08
32    1.668192e+08
18    1.551147e+08
22    1.470756e+08
12    1.442872e+08
26    1.434164e+08
34    1.382498e+08
40    1.378703e+08
35    1.315207e+08
8     1.299512e+08
17    1.277821e+08
45    1.123953e+08
21    1.081179e+08
25    1.010612e+08
```

```
43    9.056544e+07
15    8.913368e+07
7     8.159828e+07
42    7.956575e+07
9     7.778922e+07
29    7.714155e+07
16    7.425243e+07
37    7.420274e+07
30    6.271689e+07
3     5.758674e+07
38    5.515963e+07
36    5.341221e+07
5     4.547569e+07
44    4.329309e+07
33    3.716022e+07
Name: Weekly_Sales, dtype: float64
```

4 2. Monthly sales

[18]:
```python
import pandas as pd

df = pd.read_csv("Walmart_Sales.csv")

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

df['Month'] = df['Date'].dt.month

monthly_sales = df.groupby('Month')['Weekly_Sales'].sum()

print(monthly_sales)
```

```
Month
1     3.325984e+08
2     5.687279e+08
3     5.927859e+08
4     6.468598e+08
5     5.571256e+08
6     6.226299e+08
7     6.500010e+08
8     6.130902e+08
9     5.787612e+08
10    5.847848e+08
11    4.130157e+08
12    5.768386e+08
Name: Weekly_Sales, dtype: float64
```

5 3. Holiday vs non-holiday sales

[19]:
```
df.groupby('Holiday_Flag')['Weekly_Sales'].mean()
```

[19]:
```
Holiday_Flag
0    1.041256e+06
1    1.122888e+06
Name: Weekly_Sales, dtype: float64
```

6 Visulization

[21]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Walmart_Sales.csv")

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

df['Month'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year
```

[22]:
```
df
```

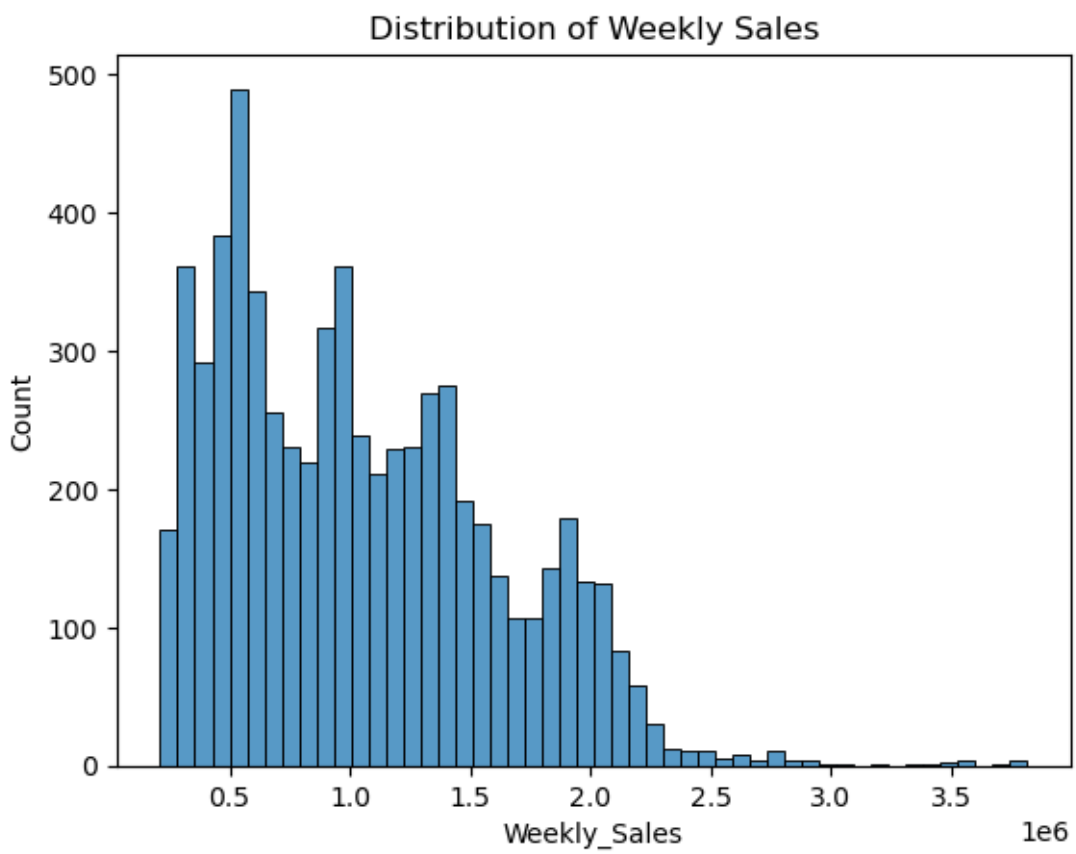
[22]:
```
      Store       Date  Weekly_Sales  Holiday_Flag  Temperature  Fuel_Price  \
0         1 2010-02-05    1643690.90             0        42.31       2.572
1         1 2010-02-12    1641957.44             1        38.51       2.548
2         1 2010-02-19    1611968.17             0        39.93       2.514
3         1 2010-02-26    1409727.59             0        46.63       2.561
4         1 2010-03-05    1554806.68             0        46.50       2.625
…       …         …            …             …            …           …
6430     45 2012-09-28     713173.95             0        64.88       3.997
6431     45 2012-10-05     733455.07             0        64.89       3.985
6432     45 2012-10-12     734464.36             0        54.47       4.000
6433     45 2012-10-19     718125.53             0        56.47       3.969
6434     45 2012-10-26     760281.43             0        58.85       3.882

             CPI  Unemployment  Month  Year
0     211.096358         8.106      2  2010
1     211.242170         8.106      2  2010
2     211.289143         8.106      2  2010
3     211.319643         8.106      2  2010
4     211.350143         8.106      3  2010
…           …             …     …    …
6430  192.013558         8.684      9  2012
6431  192.170412         8.667     10  2012
```

```
6432  192.327265          8.667      10  2012
6433  192.330854          8.667      10  2012
6434  192.308899          8.667      10  2012

[6435 rows x 10 columns]
```

7 1. Univarient Analysis

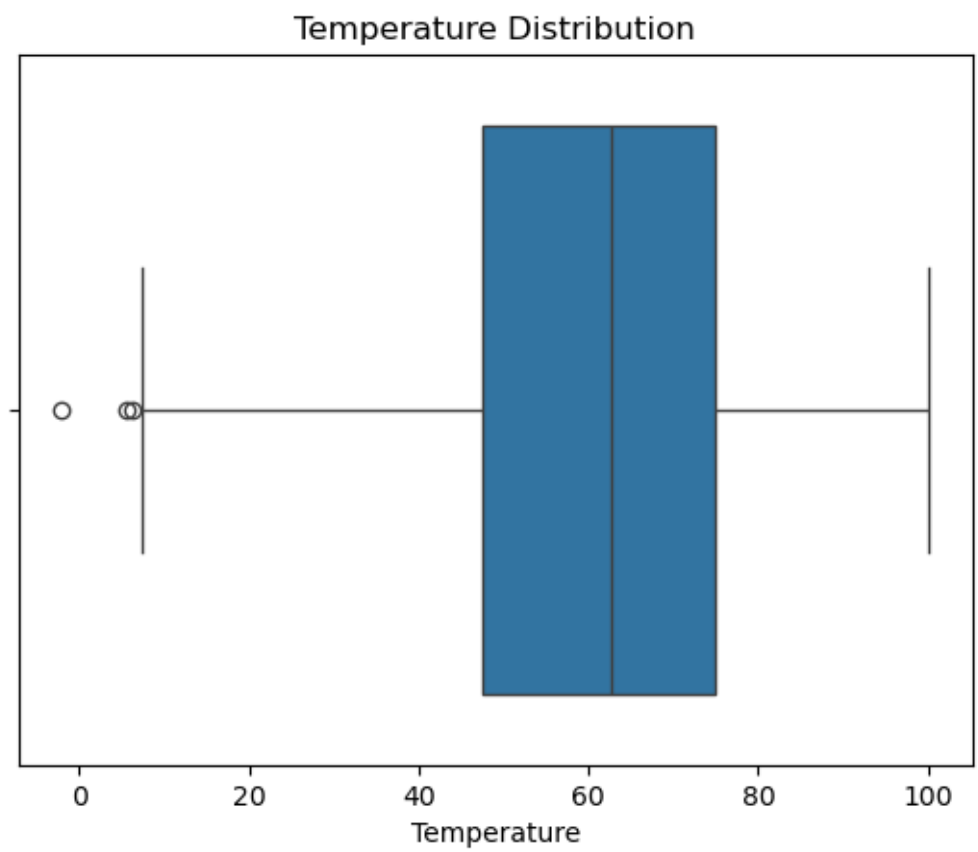
8 1. Weekly Sales Distribution

[23]:
```
sns.histplot(df['Weekly_Sales'], bins=50)
plt.title("Distribution of Weekly Sales")
plt.show()
```



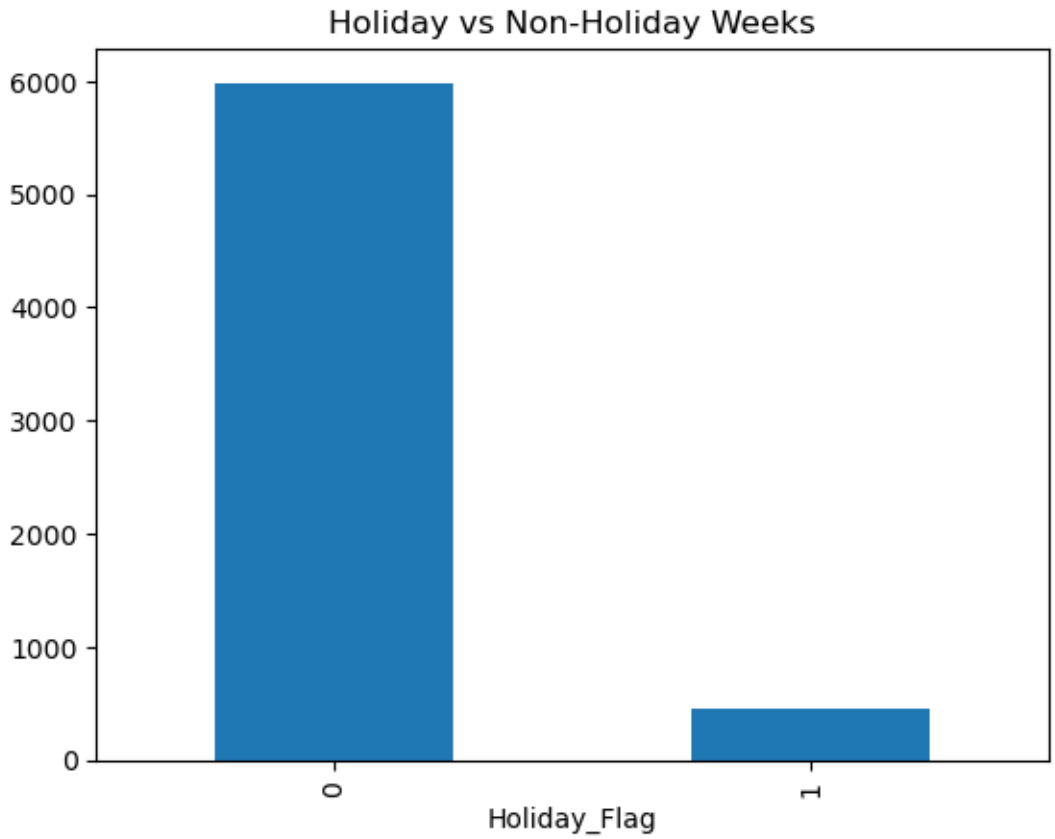
9 2. Temperature Distribution

[24]:
```
sns.boxplot(x=df['Temperature'])
plt.title("Temperature Distribution")
plt.show()
```



10 3.Holiday Frequency

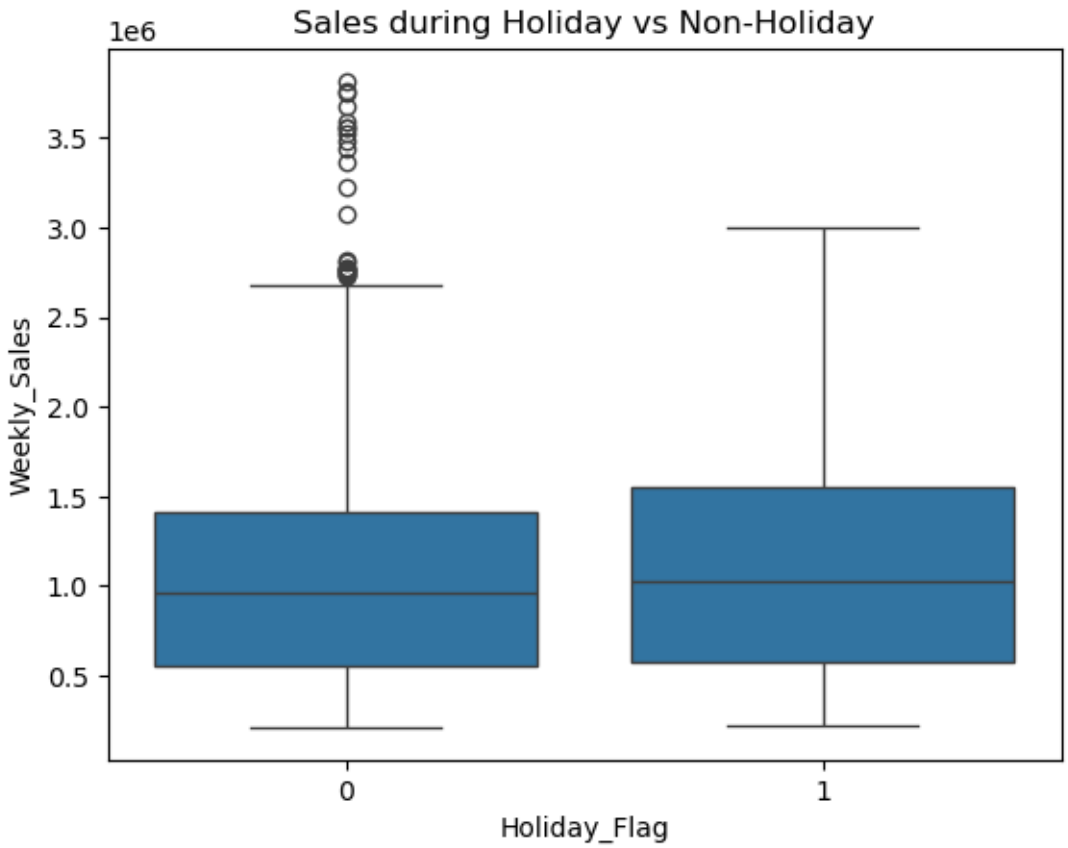
[25]:
```
df['Holiday_Flag'].value_counts().plot(kind='bar')
plt.title("Holiday vs Non-Holiday Weeks")
plt.show()
```



11 2. Bivariate Analysis (Two Variables)

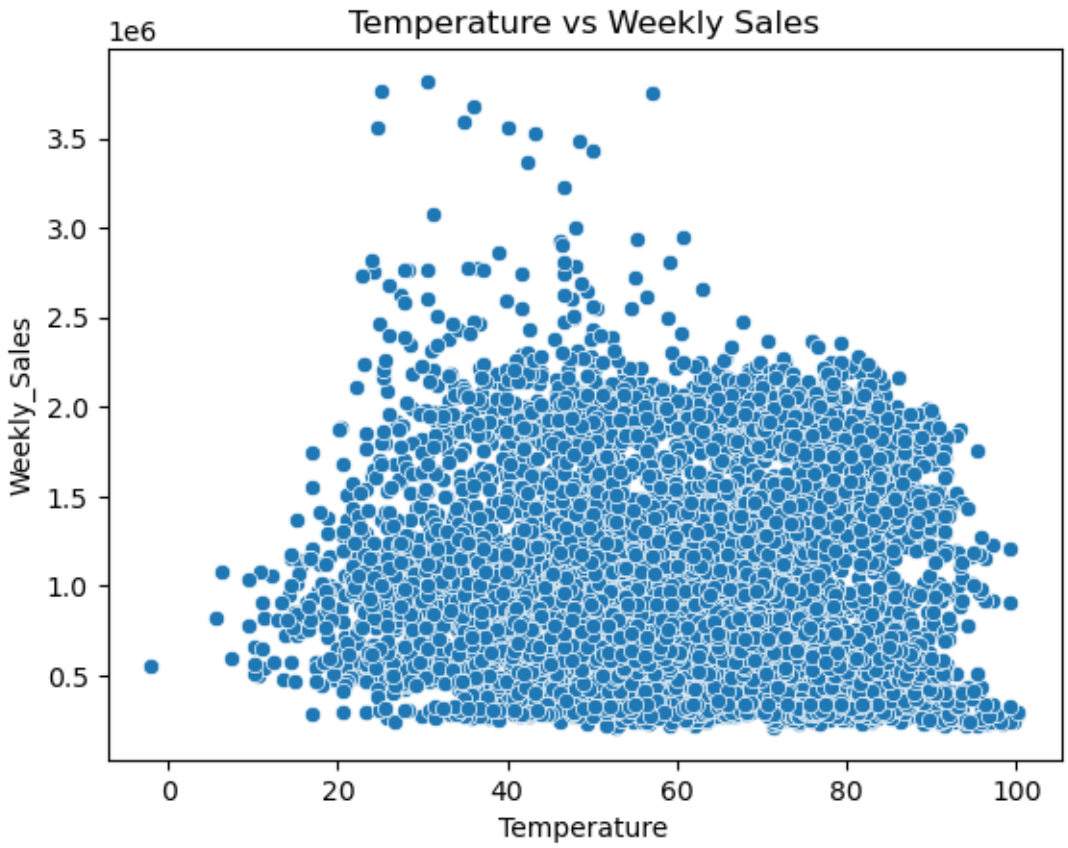
12 1. Sales vs Holiday

[26]:
```
sns.boxplot(x='Holiday_Flag', y='Weekly_Sales', data=df)
plt.title("Sales during Holiday vs Non-Holiday")
plt.show()
```



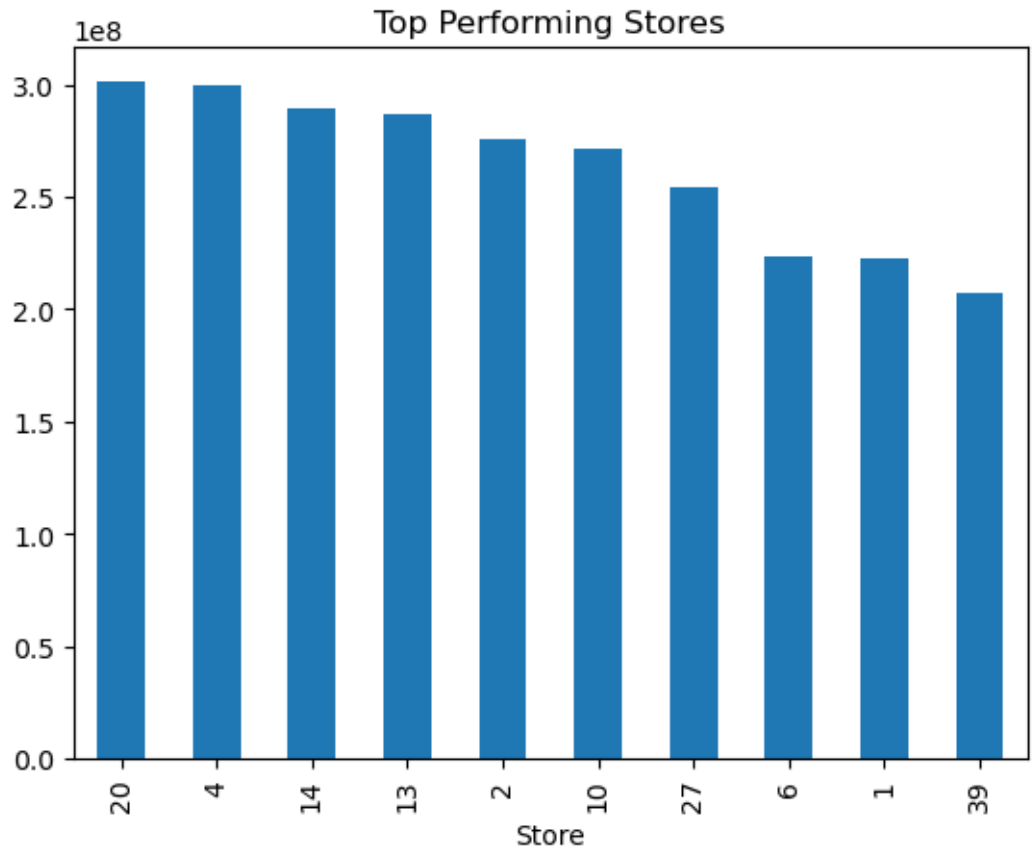
13 2. Temperature vs Sales

[27]:
```
sns.scatterplot(x='Temperature', y='Weekly_Sales', data=df)
plt.title("Temperature vs Weekly Sales")
plt.show()
```



14 3. Store Performance

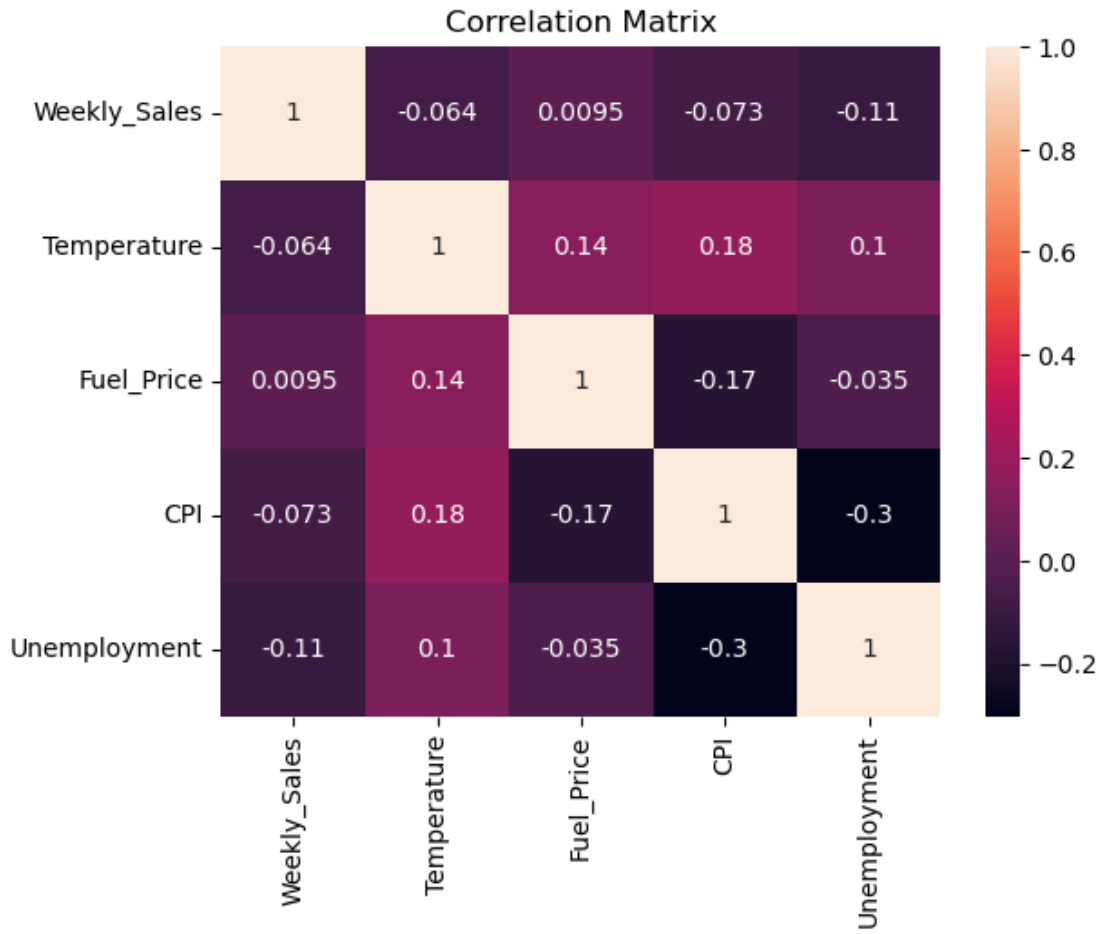
[28]:
```
store_sales = df.groupby('Store')['Weekly_Sales'].sum()

store_sales.sort_values(ascending=False).head(10).plot(kind='bar')
plt.title("Top Performing Stores")
plt.show()
```



15 3. Multivariate Analysis (Multiple Variables)

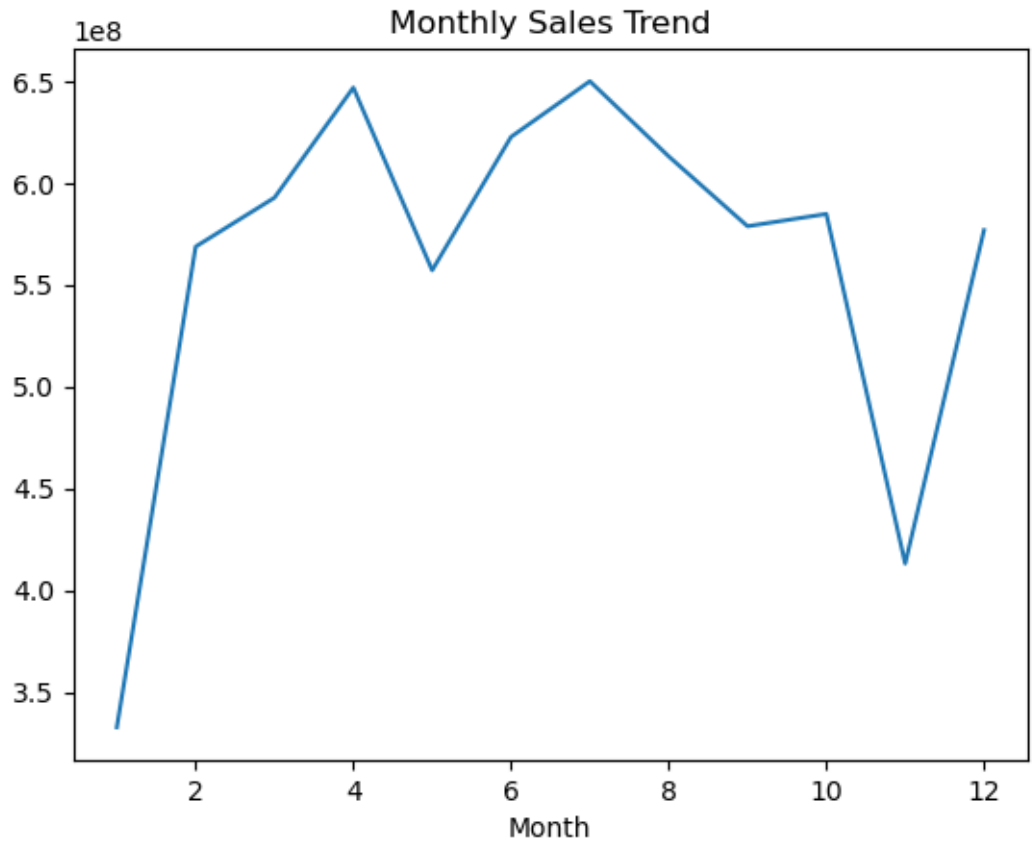
16 1. Correlation Analysis

[29]:
```
corr = df[['Weekly_Sales','Temperature','Fuel_Price','CPI','Unemployment']].
 ↪corr()

sns.heatmap(corr, annot=True)
plt.title("Correlation Matrix")
plt.show()
```



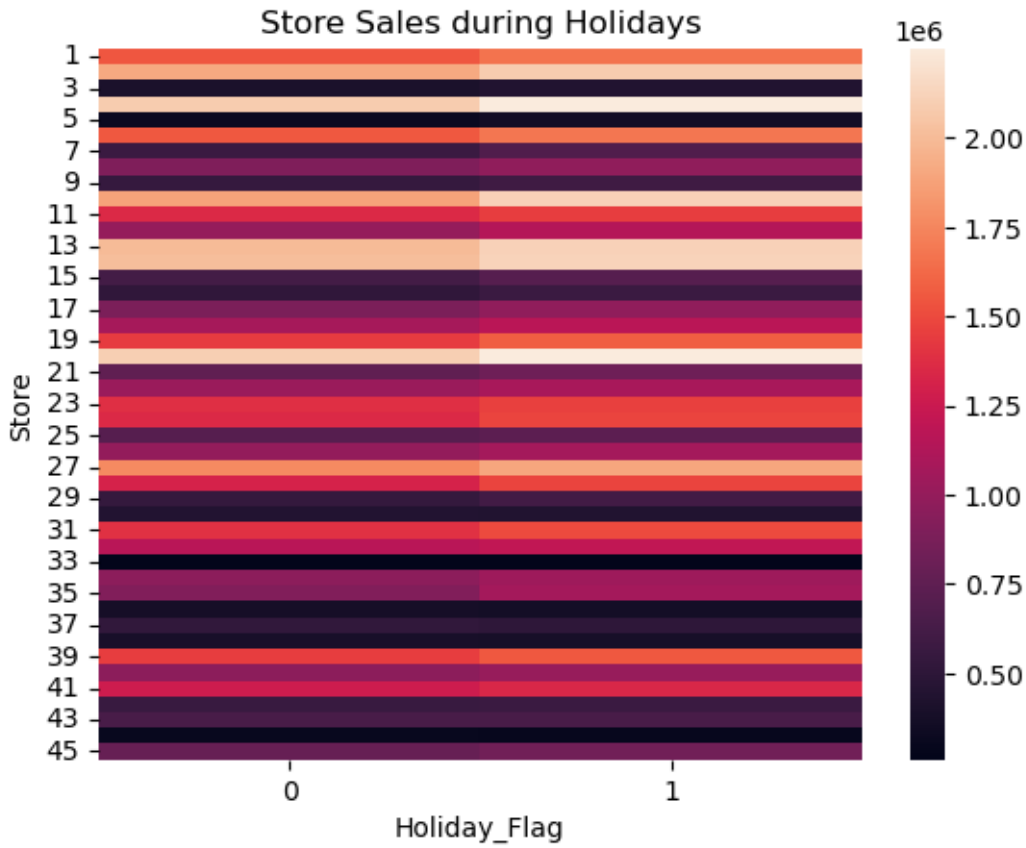
17 2. Monthly Sales Trend

[30]:
```
monthly_sales = df.groupby('Month')['Weekly_Sales'].sum()

monthly_sales.plot(kind='line')
plt.title("Monthly Sales Trend")
plt.show()
```



18 3.Sales by Store and Holiday

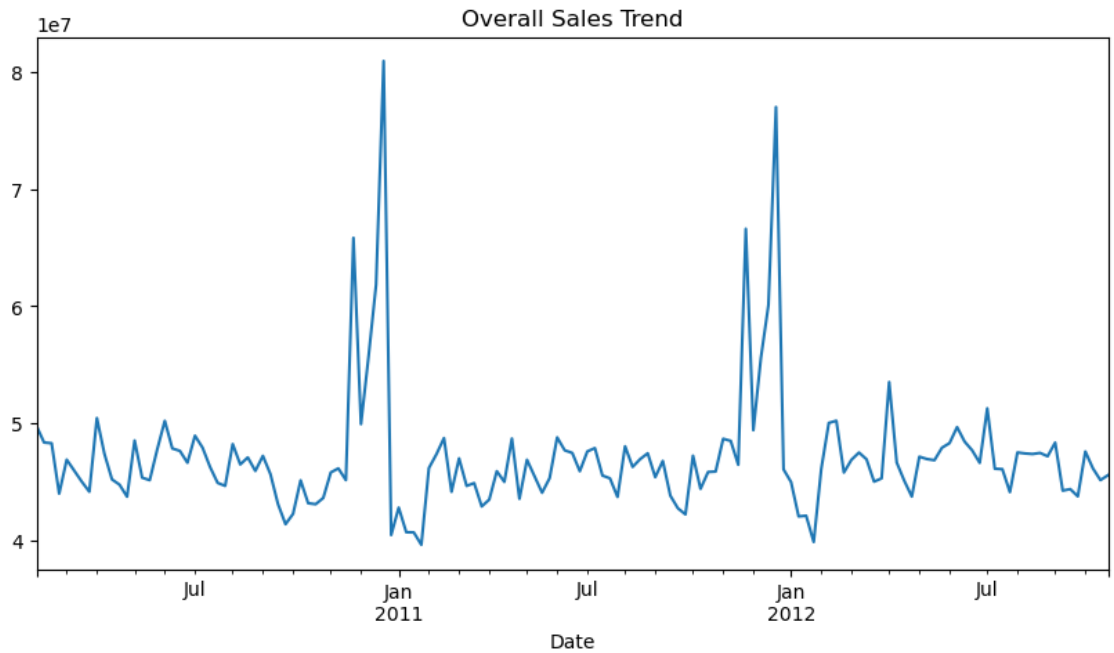
[31]:
```
pivot = pd.pivot_table(df, values='Weekly_Sales', index='Store',␣
↪columns='Holiday_Flag')

sns.heatmap(pivot)
plt.title("Store Sales during Holidays")
plt.show()
```



19 4.Advanced Industry Analysis

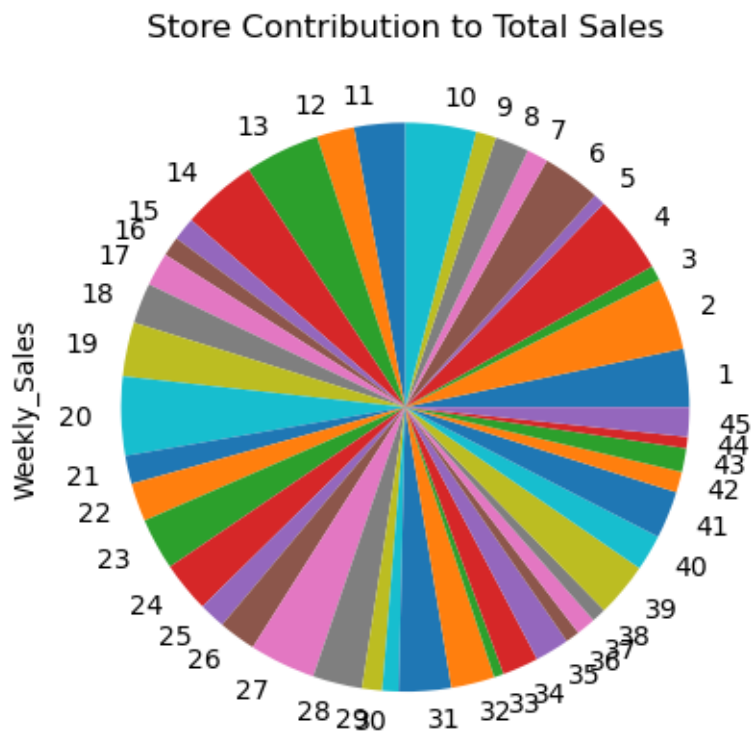
20 1.Sales Trend Over Time

[32]:
```
df.groupby('Date')['Weekly_Sales'].sum().plot(figsize=(10,5))
plt.title("Overall Sales Trend")
plt.show()
```



21 2.Store Contribution to Total Revenue

[33]:
```
store_share = df.groupby('Store')['Weekly_Sales'].sum()

store_share.plot(kind='pie')
plt.title("Store Contribution to Total Sales")
plt.show()
```



22 5.KPI Metrics

23 1.Total Sales

[34]:
```
df['Weekly_Sales'].sum()
```

[34]: 6737218987.11

24 2.Average Weekly Sales

[35]:
```
df['Weekly_Sales'].mean()
```

[35]: 1046964.8775617715

25 3.Top Store

[36]:
```
df.groupby('Store')['Weekly_Sales'].sum().idxmax()
```

[36]: 20

[ ]:

```

```